Quantifying the Effectiveness of Independent Manual-Valve Verification in Preventing Product-Transfer Risks: A Monte Carlo Process-Safety Assessment

Leon Sandler

Independent Researcher, Northbrook, Illinois, USA

ORCID: https://orcid.org/0009-0007-4584-808X

Corresponding author: Leon Sandler, sandler.leon@gmail.com

Author contributions (CRediT): Leon Sandler, sole author: conceptualization, methodology, software, formal analysis, writing – original draft, writing – review and editing.

Funding: This work received no external funding.

Declaration of competing interest: the author has prepared a related commercial proposal in response to a public open-innovation challenge concerning real-time verification of manual valve position and may benefit commercially from the concept described. The author has no other competing interests.

Declaration of generative AI and AI-assisted technologies in the writing process: a generative AI tool (Claude, Anthropic) assisted with drafting text, writing and testing simulation code and generating figures from that code. The author reviewed and edited the content as needed and takes full responsibility for the content.

Data availability: the code, the data of every figure and table, the tests and the manuscript are archived at https://doi.org/10.5281/zenodo.24046773 (software, all versions) and https://doi.org/10.5281/zenodo.24046771 (manuscript, all versions), and are available at https://github.com/sandlerleon/hydrasense-valve-verification-mc.